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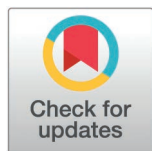
Plasma proteomic signatures of early retinal neurodegeneration in diabetes: a multi-cohort study

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Abstract

Background

Retinal neurodegeneration is an early and independent feature of diabetic retinal disease and has been proposed as a window into the systemic neural consequences of diabetes, yet accessible molecular biomarkers and individualized prediction tools remain scarce. We aimed to identify circulating plasma protein signatures of diabetic retinal neurodegeneration (DRN) and to translate them into a clinically usable risk prediction system.

Methods and findings

In this multi-cohort prospective observational study, we integrated high-throughput plasma proteomics with longitudinal optical coherence tomography (OCT) in two independent populations. The discovery cohort comprised 1,492 participants had baseline plasma proteomics and OCT, and 1,218 were followed with repeated OCT over 6 years in Guangzhou Diabetic Eye Study (GDES). DRN was quantified by the annualized OCT-derived retinal nerve fiber layer thinning rate. In multivariable analyses adjusted for age, sex, smoking, systolic blood pressure, HbA1c, and diabetes duration, we

Center, Sun Yat-sen University. Requests for access should be directed to the data manager via email at liyuting@gzzoc.com. Requests will be reviewed within 90 days according to Center policy, and, if approved, an inter-institutional data use agreement specifying use for non-commercial academic research will be required. UK Biobank data are available through application via the official platform (<http://www.ukbiobank.ac.uk>) under a material transfer agreement (application number: 105658). All software used in this study is publicly available. The code used in this study can be accessed at <https://github.com/ZOC-skl/Pro-DRN>.

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Abbreviations: BCVA, best corrected visual acuity; BRB, blood–retina barrier; COIP, Chinese Ocular Imaging Project; DCA, decision curve analysis; DR, diabetic retinopathy; DRD, diabetic retinal disease; DRN, diabetic retinal neurodegeneration; GDES, Guangzhou Diabetic Eye Study; GO, Gene Ontology; IOP, intraocular pressure; KEGG, Kyoto Encyclopedia of Genes and Genomes; KNN, k-nearest neighbors; mfERG, multifocal electroretinography; ML, machine learning; MNSI, Michigan Neuropathy Screening Instrument; NPX, Normalized Protein eXpression; OCT, optical coherence tomography; PEA, proximity extension assay; Pro-DRN, proteomics-based DRN model; RNFL, retinal

identified 71 plasma proteins associated with development and progression of DRN. These proteins mapped onto pathways governing inflammatory immune recruitment, extracellular matrix remodeling, and microvascular homeostasis, providing a plausible biological basis for DRN. We developed a proteomics-based DRN model (Pro-DRN) using eight machine learning (ML) algorithms, including XGBoost and LightGBM. In the independent test set, Pro-DRN achieved a C-index of 0.860, rising to 0.908 when integrated with clinical variables. Compared with six conventional models, Pro-DRN improved discrimination (Δ C-index 0.137 to 0.159; all $P < 0.001$), reclassification (IDI 0.212 to 0.245; NRI 0.226 to 0.452; all $P < 0.05$). In the Hippiisley model, the C-index increased from 0.739 (95% CI [0.670, 0.808]) to 0.898 (95% CI [0.858, 0.937]), with IDI 0.245 (95% CI [0.177, 0.318]), NRI 0.452 (95% CI [0.222, 0.673]) (both $P < 0.001$), and higher net benefit. The proteins most consistently driving model performance included ACTA2, COL6A3, and HSPG2. For clinical translation, we deployed the locked model as an interactive, web-based risk-assessment tool to support early DRN screening and longitudinal monitoring. Cross-ethnic external validation in UK Biobank ($n = 502$; recruited 2006–2010) reproduced core protein signals and consistent effect directions, confirming robustness across populations. Principal methodological limitation lies in single time point proteomic assessment.

Conclusion

In this multi-cohort study, we present a proteomics- and ML-based precision prediction system for DRN. Pro-DRN substantially enhanced early risk stratification beyond conventional clinical factors and may support targeted screening and timely neuro-protective interventions, advancing molecularly guided strategies for diabetic eye disease prevention.

Author summary

Why was this study done?

- Diabetic retinal neurodegeneration is an early form of diabetes-related damage to the light-sensing layer of the eye, and growing evidence suggests that it also mirrors wider nerve injury elsewhere in the body; despite this, no practical tool exists to identify which people with diabetes are at highest risk before irreversible damage has occurred.
- Current screening relies on standard clinical risk factors and retinal photographs, both of which tend to detect damage only after it has already developed, leaving a narrow window for prevention.
- High-throughput plasma proteomics, combined with explainable AI, may offer a scalable way to identify blood-based signals of DRN and translate them into clinically useful prediction tools.

nerve fiber layer; SHAP, SHapley additive explanations; SVM, support vector machine; TABS, Topcon Advanced Boundary Segmentation; UKB, UK Biobank.

What did the researchers do and find?

- We analyzed two large prospective cohorts containing both high-resolution retinal imaging and measurements of thousands of blood proteins, using the Guangzhou Diabetic Eye Study for discovery ($n = 1,492$) and model development, and UK Biobank ($n = 502$) as an external validation cohort.
- We identified a group of blood proteins linked to DRN and developed a prediction model (Pro-DRN) based on these protein signals.
- To support clinical translation, we implemented the final model as an interactive online tool for real-time DRN risk assessment.

What do these findings mean?

- These findings suggest that a simple blood test analyzed with artificial intelligence may help identify people with diabetes who are at highest risk of early retinal nerve damage, well before visible damage appears on the retina.
- The results also reinforce the emerging view that retinal neurodegeneration reflects broader nerve and blood vessel injury in diabetes, so that a single eye-based test may carry information about the wider health of the nervous system.
- Important limitations should be kept in mind: the protein signals were measured in blood rather than directly in the retina, samples were obtained at a single time point, and the findings describe statistical associations rather than proven causes.

Introduction

Diabetes now affects more than half a billion adults worldwide, and the burden of visual impairment and blindness attributable to diabetic eye disease has remained stubbornly high across three decades of clinical progress [1–3]. Traditionally viewed as a microvascular disorder, diabetic retinal disease (DRD) is now recognized as a highly tissue-specific neurovascular complication [3–5]. Diabetic retinal neurodegeneration (DRN) often precedes microvascular pathology, independently driving visual dysfunction and disease progression [3,6,7]. Furthermore, DRN is regarded as a “window” into central nervous system complications of diabetes, such as cognitive impairment, dementia, and peripheral neuropathy [8–12].

Current screening for DRD relies on demographics, metabolic markers, and fundus photography, none of which can reliably flag individuals at highest risk of early neurodegeneration [3]. Advances in retinal imaging, including optical coherence tomography (OCT) and multifocal electroretinography (mfERG), enable in vivo assessment of neurodegeneration, yet they predominantly capture late structural or functional changes and fail to reflect underlying molecular processes [8–13]. By the time these structural or functional deficits become measurable, neurodegeneration is

often irreversible, creating an urgent need for accessible, interpretable, blood-based biomarkers that provide early warning while neural function remains plastic and amenable to intervention.

Circulating plasma proteins, as dynamic executors of organ crosstalk, offer a unique opportunity to bridge this diagnostic gap [14–16]. Their stability and measurability confer high translational potential. Previous proteomic studies of DRD have focused overwhelmingly on vascular leakage and intraocular fluids, leaving neurodegeneration-relevant signatures in the systemic circulation largely unexplored [17]. Advances in high-throughput plasma proteomics, such as Olink proximity extension assay, now enable precise quantification of thousands of proteins, providing the technical foundation for large-scale biomarker discovery. Integrating proteomics with longitudinal retinal imaging offers a path to systematically identify molecular drivers of DRN, detect high-risk individuals early, and reveal intervention targets [18,19].

The high dimensionality, nonlinearity, and complex interaction structure of modern proteomic data exceed the capacity of conventional statistical approaches. Multi-algorithm machine learning (ML), combined with explainable artificial intelligence, now enables robust signal extraction alongside interpretable biological insight [14]. Explainable AI methods, such as SHapley additive explanations (SHAP), map protein contributions, linking predictive performance to clinical and mechanistic insight. Integrated ML strategies combining proteomics, imaging, and clinical variables have shown remarkable promise for early diagnosis and risk stratification across diabetes, dementia, cardiovascular disease, and mortality [20–28]. To date, DRN biomarker research faces critical gaps [17,29–31]: (1) most studies are cross-sectional, limiting causal inference and predictive validation; (2) previous studies focus on late structural or functional changes, with little longitudinal data capturing early neurodegeneration; (3) high-throughput proteomics is rarely integrated with imaging and clinical data in a unified framework; (4) biomarkers often lack cross-cohort validation and generalizability; and (5) interpretable, blood-based panels suitable for clinical workflows are largely absent, constraining early risk identification and targeted intervention.

To address these challenges, we leveraged two large prospective cohorts, the Guangzhou Diabetic Eye Study (GDES) and the UK Biobank (UKB), to test the hypothesis that circulating plasma proteins carry early, quantifiable signatures of DRN that can be translated into a clinically usable risk prediction tool. We quantified DRN using longitudinal OCT, integrated high-throughput plasma proteomics with multi-algorithm ML frameworks, and pursued four prespecified objectives: (i) to identify plasma protein signatures associated with baseline and longitudinal retinal nerve fiber layer (RNFL) thinning; (ii) to replicate core findings across ethnicities in UKB; (iii) to elucidate biological pathways linking these proteins to neurovascular instability; and (iv) to construct, interpret, and externally evaluate a proteomics-based DRN prediction system suitable for early risk stratification and targeted prevention.

Methods

Study design and participants

This international multi-cohort prospective observational study was designed to identify plasma protein biomarkers of DRN and to develop, interpret, and externally validate a machine learning based prediction model suitable for clinical translation. Two large-scale prospective cohorts were used: the Guangzhou Diabetic Eye Study (GDES) as the discovery and training cohort, and the UKB as the external validation cohort [32–35]. GDES is a large prospective cohort of adults with type 2 diabetes integrating plasma proteomics and multimodal retinal imaging. Between 2017 and 2019, over 3,000 individuals with type 2 diabetes aged 35–85 years were recruited in Guangzhou, China. Its proteomics sub-cohort (GDES-PPP) served as the development dataset. UKB is a population-based multicenter cohort that enrolled over 500,000 participants aged 40–69 years across the United Kingdom between 2006 and 2010. Its proteomics subproject (UKB-PPP) was used for external validation. Both studies adhered to the Declaration of Helsinki. Ethical approvals were obtained from the Zhongshan Ophthalmic Center Ethics Committee (2017KYPJ094) and the North West Multi-Centre Research Ethics Committee (11/NW/0382). All participants provided written informed consent. The study was reported following STROBE guidelines (S1 Checklist) and TRIPOD statement (S2 Checklist).

Participants from GDES were included in two steps. First, Population 1 comprised 1,492 participants with type 2 diabetes but no diabetic retinopathy (DR) at baseline, all with baseline proteomics and qualified OCT data, and was used for cross-sectional analysis of plasma protein associations with retinal nerve fiber layer thickness. Second, to focus on neurodegeneration independent of vascular or exudative lesions, we identified within Population 1 a subgroup of 1,218 participants who remained free of DR during 6 years of follow-up (Population 2). All individuals in Population 2 had baseline proteomic data and OCT scans, with OCT repeated at years 2, 4, and 6 after enrollment. This population was used for longitudinal RNFL thinning analyses, pathway enrichment, and ML model development. For external validation, Population 3 comprised 502 UKB participants with type 2 diabetes but no baseline DR ([S1 Table](#)), all with available proteomics and a single time point OCT measure obtained at the UKB baseline ocular imaging assessment (2009–2010). Across both cohorts, we excluded participants lacking plasma proteomics or analyzable OCT-derived RNFL measures, applied prespecified OCT quality control, and excluded glaucoma, dementia, and other ocular conditions that could confound RNFL measurements ([S1 Text](#)). This cohort was used for cross-sectional analyses of associations between baseline protein levels and RNFL thickness to evaluate the robustness and generalizability of protein associations across diverse ethnicities and healthcare systems. The prespecified Phase I–IV objectives are summarized in the Results.

Retinal OCT imaging protocols

High-resolution OCT was used to quantify RNFL thickness, a key biomarker of DRN [[36–39](#)]. Standardized acquisition and analysis procedures were applied in both cohorts: (1) GDES cohort: Disc-centered three-dimensional OCT scans were acquired under dark-room conditions using the Topcon DRI OCT Triton swept-source system (Topcon Corporation, Japan), featuring a 1,050 nm light source, 100,000 A-scans/s, and 8 μ m axial resolution. Peripapillary RNFL thickness was measured along a 360° circular path, 3.4 mm in diameter centered on the optic disc. (2) UKB cohort: Three-dimensional volume scans were obtained using the Topcon 3D OCT-1000 Mk II (Topcon Corporation, Japan) under similar dark-room conditions, with 18,000 A-scans/s and 6 μ m axial resolution.

In GDES cohorts, scans were performed after adequate mydriasis by trained technicians, with internal eye-tracking to ensure accuracy. In UK Biobank, OCT imaging followed the standard UKB protocol and was performed without pharmacologic mydriasis. RNFL segmentation was performed automatically using the Topcon Advanced Boundary Segmentation (TABS) algorithm, defining RNFL thickness as the perpendicular distance from the internal limiting membrane to the inner border of the ganglion cell layer. The peripapillary region was divided into four quadrants (superior, inferior, temporal, and nasal), and both average and quadrant-specific RNFL thicknesses were calculated. All images were visually inspected by two experienced OCT technicians, with manual correction applied as needed. Only high-quality scans were included; images with poor quality, motion artifacts, or signal strength index <60 were excluded. For participants with valid data in both eyes, right-eye measurements were preferentially used ([S1 Text](#)).

DRN definition and phenotype quantification

DRN was defined as a pathological state marked by accelerated RNFL loss. Using repeated OCT measurements at baseline, year 2, year 4, and year 6, the annualized RNFL thinning rate was estimated using a linear mixed-effects model (median follow-up, 5.49 years). In the GDES cohort, participants were ranked by this rate: those in the fastest quartile (Q1) were classified as DRN, while the remaining participants (Q2–Q4) served as non-DRN controls. This objective, biologically grounded definition identifies individuals with pronounced neurodegenerative phenotypes, providing a precise foundation for biomarker discovery and predictive modeling. In sensitivity analyses, we additionally defined DRN as (i) the fastest 10% of RNFL thinners (DRN-Top10Slope), (ii) thin/abnormal pRNFL on the OCT report at the final available visit (DRN-ThinLast), and (iii) an annualized RNFL thinning rate below the lower normative reference limit derived from the 6-year longitudinal OCT data of healthy participants in the Chinese Ocular Imaging Project (COIP) cohort (DRN-ExcessLoss; [S1 Text](#)).

Plasma proteomics protocols

High-throughput plasma proteomics was performed using Olink Proximity Extension Assay (PEA) technology in both GDES and UKB cohorts, following standardized protocols for sample collection, preprocessing, measurement, and quality control. Laboratory personnel were blinded to clinical data. In GDES, a targeted approach employed the Olink Cardiometabolic Panel, measuring 361 proteins relevant to metabolic regulation, cardiovascular function, and inflammatory pathways, all mechanistically plausible for diabetes and its neuro-complications. This focused panel enabled deep, pathway-specific interrogation of DRN biology. The UKB cohort applied a broader proteomic panel, providing an independent platform for validation and assessment of biomarker robustness and generalizability. All proteins assayed in GDES were included in the UKB-PPP protein set; therefore, all cross-cohort replication and validation were conducted using this shared set, which comprised the full GDES panel. Rigorous quality control was implemented in both cohorts: within-plate CV <10% and between-plate CV <20%, with inter-plate normalization using control samples. Raw counts were normalized by extension controls and log₂-transformed to generate Normalized Protein eXpression (NPX) values, minimizing technical variation [26,27]. All NPX data were standardized and batch-corrected for downstream statistical analyses (S1 Text).

Covariate measurements

Covariates were collected using rigorous, standardized procedures within the prospective GDES cohort. Data encompassed demographics, lifestyle factors, clinical indices, and ophthalmic parameters, ensuring accuracy and comparability. Demographic (age, sex), socioeconomic (education, household income), and lifestyle data (smoking, alcohol, physical activity) were obtained through validated electronic questionnaires combined with face-to-face interviews. Diabetes-related information, including duration, medication use, and complications, was ascertained via medical record review and patient interview, with diagnoses based on international criteria and independently confirmed by two internists.

Anthropometric and biochemical measurements were performed by trained staff using standardized protocols. Height and weight were measured with calibrated instruments; blood pressure was recorded on the non-dominant arm after 5 min of rest, with three consecutive measures averaged. Fasting venous blood was collected for HbA1c (HPLC), lipid panel (total cholesterol, triglycerides, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol via enzymatic colorimetry), renal function markers (serum creatinine, estimated glomerular filtration rate), serum urate, and urinary albumin. All assays incorporated internal quality-control samples and participated in external quality assessment programs.

Ophthalmic evaluations followed standardized protocols. Slit-lamp biomicroscopy (Haag-Streit BQ-900, Switzerland) assessed the anterior segment. Best corrected visual acuity (BCVA) was measured using Early Treatment Diabetic Retinopathy Study (ETDRS) LogMAR charts (Precision Vision, USA) at 4 m. Refractive status was measured with an autorefractor (Topcon KR-8800, Japan); intraocular pressure (IOP) was measured with a non-contact tonometer (Topcon CT-1, Japan); axial length was measured by optical biometry (Lenstar LS900, Haag-Streit, Switzerland). All assessments were conducted by trained personnel following uniform procedures. Baseline covariate protocols for the UKB cohort are detailed in S1 Text.

Functional enrichment analysis

To elucidate the biological mechanisms underlying plasma proteins associated with DRN, we conducted comprehensive functional enrichment analyses. Using the set of plasma protein markers significantly linked to DRN, we applied the clusterProfiler R package (v4.0) to perform enrichment against Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), and Reactome databases. Analyses encompassed biological process (BP), cellular component (CC), and molecular function (MF) terms, KEGG pathways, and Reactome reaction networks, systematically profiling the biological activities, subcellular localization, molecular functions, and pathway interactions of the target proteins. Enrichment

significance was assessed using the hypergeometric (Fisher's) test, with P values adjusted for multiple comparisons by the Benjamini–Hochberg method (FDR) < 0.05 was considered statistically significant.

Pro-DRN model construction

To develop a proteomics-driven risk prediction model, we selected plasma proteins significantly associated with longitudinal RNFL thinning in the discovery stage as features. The GDES cohort was randomly split into training and test sets at an 8:2 ratio, stratified by DRN status to preserve the proportion of positive cases. All data preprocessing, feature selection, and hyperparameter tuning were confined to the training set; the test set was used solely for final evaluation to prevent information leakage.

To ensure robustness and cross-validate the proteomic signal, we trained eight supervised learning algorithms in parallel: XGBoost, LightGBM, Random Forest, support vector machine (SVM), Neural Network, Logistic Regression, k-nearest neighbors (KNN), and Decision Tree. Stratified 10-fold cross-validation was applied within the training set, using fold-averaged C-index as the primary metric. Simpler models were favored when discrimination was comparable, and in-fold calibration was evaluated. Gradient-boosting models incorporated early stopping to mitigate overfitting, and hyperparameters were optimized via grid search or Bayesian methods based on cross-validation performance ([S1 Text](#)).

The final model was chosen based on consistent superiority across cross-validation and the independent test set. Model interpretability was assessed using Shapley values, quantifying each protein's contribution to predictions. Two metrics were computed: (1) global feature importance based on mean absolute SHAP values, reflecting overall contribution, and (2) individual-level feature attributions, indicating the direction and magnitude of each protein's effect on specific predictions.

Clinically meaningful outcomes for external validity

To assess clinical relevance beyond the OCT-based DRN phenotype, we examined associations between the baseline proteomic risk score and subsequent vision impairment and diabetic peripheral neuropathy. Vision impairment was defined as a loss of ≥ 15 ETDRS letters in best-corrected visual acuity during follow-up. Diabetic peripheral neuropathy was defined according to a previously published approach as a dichotomized Michigan Neuropathy Screening Instrument (MNSI) patient questionnaire score ≥ 4 .

Statistical analysis

Statistical analysis protocol is provided in the Supplementary Materials. Analyses were conducted using R (v4.5.1) and Stata/MP (v18.0). Continuous variables are presented as mean $\pm$ SD or median (IQR), and categorical variables as counts (percentages). Between-group comparisons used t -tests or χ^2 tests as appropriate. All tests were two-sided, with $P < 0.05$ considered statistically significant. Multiple comparisons were controlled using the Benjamini–Hochberg method, with FDR < 0.05 considered significant.

In Population 1, cross-sectional associations between plasma proteins and RNFL thickness were assessed using multivariable linear regression, adjusting for age, sex, smoking, systolic blood pressure, HbA1c, and diabetes duration. Proteins with FDR < 0.05 were deemed significantly associated. In Population 2, the same covariates were used to test longitudinal associations with RNFL thinning rate.

For DRN prediction, participants in the top 25% of RNFL decline were classified as DRN cases, with the remainder as controls. Samples were randomly split 8:2 into training and testing sets. Using selected protein features, Pro-DRN were built with eight ML algorithms. The optimal model was selected based on C-index and generalization performance. Model evaluation included comparisons with nine individual traditional predictors and a baseline age–sex model using C-index and ΔC as discrimination metrics. Pro-DRN was further integrated into multiple existing clinical prediction models (Aspelund, Hippisley, Dagliati, ISDR, JDC, Tarasewicz, and a combined “All” model; variable composition was summarized in

[S13 Table](#)) to assess incremental predictive value [40–48]. Risk reclassification was quantified using NRI and IDI, calibration assessed with calibration plots. Decision curve analysis (DCA) was used to evaluate clinical utility, and the net-benefit definition and underlying assumptions are described in [S1 Text](#). The final model was implemented as an interactive online tool using the R Shiny [49].

For external validation, UKB-PPP participants with diabetes and available proteomics and OCT data were analyzed as an independent cohort. Cross-sectional analyses were fully replicated using the same covariates and modeling strategy to assess consistency in effect direction and significance of key protein associations. We additionally validated the locked Pro-DRN model using a cross-sectional thin-RNFL outcome, reporting discrimination (C-index) and calibration (Brier score and Hosmer–Lemeshow test).

Results

Study design and framework

This study leveraged two large-scale prospective cohorts: the Guangzhou Diabetic Eye Study (GDES) and the UK Biobank (UKB) [32–35]. The GDES Proteomic Program (GDES-PPP) served as the discovery and model development dataset, and the UKB Proteomic Program (UKB-PPP) provided external validation. Participants with both high-quality Olink proteomic data and RNFL measurements by OCT were included.

The analytical framework comprised four modules ([Fig 1](#)). Module I involved cross-sectional protein screening using GDES-PPP baseline data to identify plasma proteins associated with DRN. Module II performed longitudinal analyses among participants who remained DRD-free over six years, evaluating associations between candidate proteins and RNFL thinning rates, and revealing underlying biological pathways via pathways enrichment analyses. Module III built and evaluated ML-based DRN prediction models, assessing performance through interpretability metrics, predictive accuracy, and comparison with conventional models [49]. Module IV validated DRN-associated protein signatures in UKB participants with type 2 diabetes to confirm generalizability across populations.

Baseline characteristics

The baseline characteristics of the three analytical populations are summarized in [Tables 1](#) (Population 2) and [S1](#) (Populations 1 and 3). In GDES-PPP, 1,492 participants with type 2 diabetes, with baseline plasma proteomics and qualified macular OCT scans (Population 1) were included in cross-sectional analyses (mean age 64.4 ± 7.5 years; 57.3% women). A total of 1,218 DRD-free over six years participants were included in longitudinal analyses (Population 2). Based on quartiles of RNFL thinning rate, the fastest-declining quartile (Q1, $n=305$, 25.0%) defined the DRN group, and Q2–Q4 ($n=913$) the non-DRN group. Participants in the DRN group were older, had a longer duration of diabetes, and higher systolic blood pressure at baseline ([Table 1](#)). Baseline pRNFL thickness was comparable between DRN progressors and non-DRN participants (110.07 ± 13.67 μm versus 109.81 ± 12.29 μm ; $P=0.756$). For external validation, 502 UKB-PPP participants with type 2 diabetes, no DR or other retinal disease at the time of OCT imaging, and available proteomic and OCT data (mean age 58.3 ± 7.6 years; 50.8% women) provided an independent cross-ethnic dataset ([S1 Table](#)).

Plasma proteins associated with DRN

In the GDES-PPP discovery cohort, cross-sectional analyses identified 72 plasma proteins significantly associated with average RNFL thickness ($\text{FDR} < 0.05$; [Fig 2A](#) and [2B](#)). Among these, the 10 proteins most strongly negatively correlated with RNFL thickness were TFF3, MCFD2, CST3, FAM3C, CGREF1, NECTIN2, NPDC1, CLC, HSPG2, and DEFA1, with higher protein levels associated with thinner baseline RNFL ([S2 Table](#)). The adjusted effect sizes (β) ranging from -2.014 μm (95% CI $[-2.800, -1.229]$) to -0.983 μm (95% CI $[-1.702, -0.265]$) per 1-SD increase. In longitudinal analyses, 71 of these proteins were confirmed being significantly associated with RNFL thinning rate ($\text{FDR} < 0.05$; [S3 Table](#)). The top 10

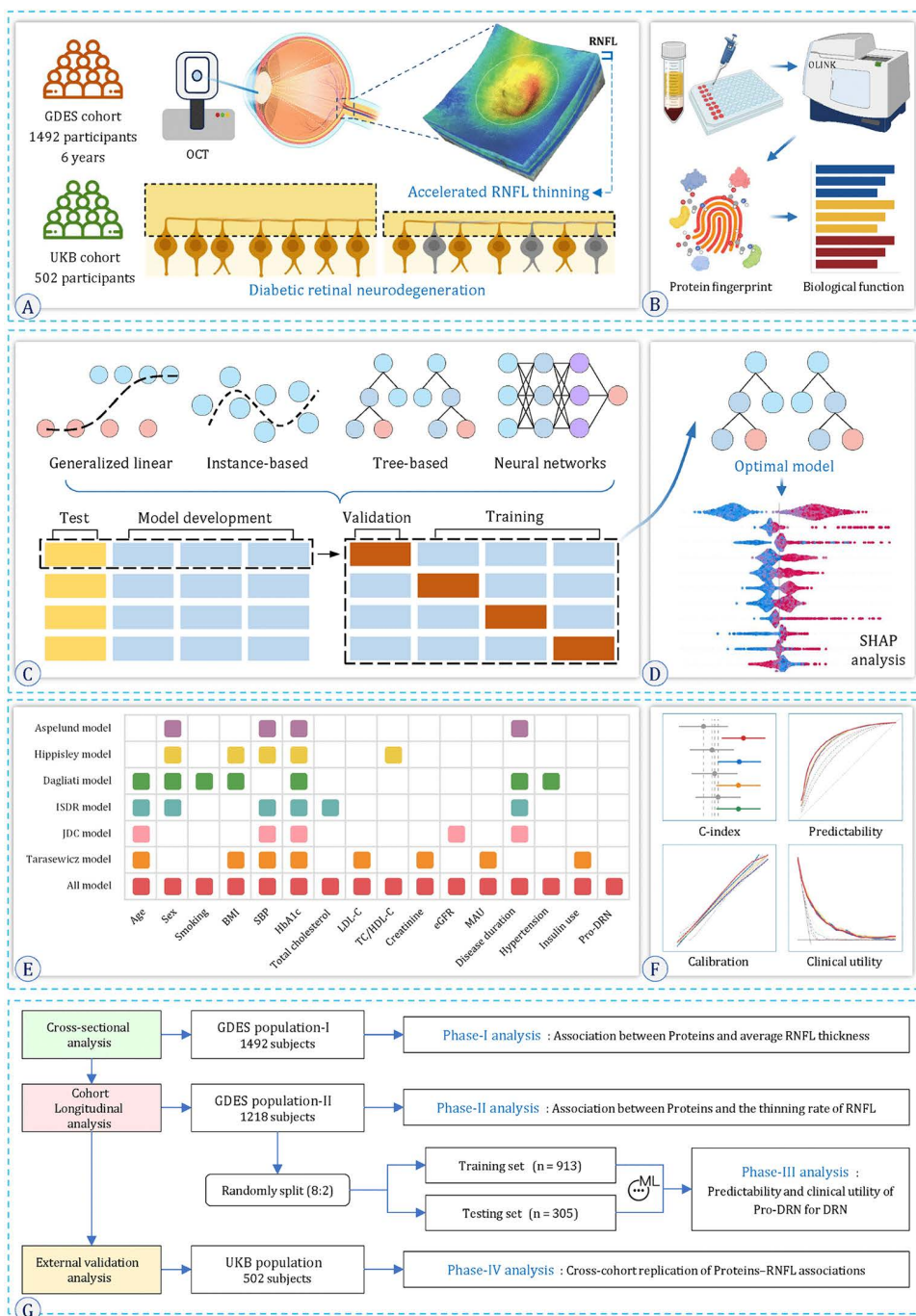


Fig 1. Study design and workflow. (A) Cohorts and phenotype. Participants from the Guangzhou Diabetic Eye Study (GDES; discovery cohort with 6-year follow-up; $n = 1,492$ baseline) and the UK Biobank (UKB; external replication; $n = 502$) underwent optical coherence tomography (OCT) to quantify retinal nerve fiber layer (RNFL) thickness; accelerated RNFL thinning operationalizes DRN. (B) Plasma proteomics and functional annotation. EDTA plasma was profiled using a proximity-extension assay (Olink) to derive protein fingerprints and functional enrichment. (C) Model development. GDES was randomly split (8:2) into training and an untouched test set. Four families of learners were trained (generalized-linear, instance-based, tree-based, and neural-network models), with internal validation for hyperparameter tuning. (D) Model selection and interpretability. The optimal model was chosen based on validation performance and interpreted with SHAP (Shapley additive explanations) to quantify feature contributions. (E) Clinical frameworks. Candidate clinical variables were organized according to established risk models (Aspelund, Hippisley, Dagliati, ISDR, JDC, Tarasewicz) and a comprehensive "All model." (F) Performance assessment. Discrimination (C-index/ROC), calibration, and clinical utility (decision-curve analysis) were evaluated

in the hold-out test set. **(G)** Analysis phases. Phase I: cross-sectional protein–RNFL thickness associations; Phase II: longitudinal associations with RNFL thinning rate; Phase III: development and clinical evaluation of Pro-DRN in GDES; Phase IV: cross-cohort replication in UKB. **Abbreviations:** RNFL, retinal nerve fiber layer; OCT, optical coherence tomography; SHAP, Shapley additive explanations.

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Table 1. Baseline characteristics of Guangzhou Diabetic Eye Study (GDES) participants with Non-DR during follow-up.

Characteristics	Overall	Non-DRN	DRN	P value*
No. of subjects	1,218	913	305	
Age, year	64.06 ± 7.34	63.36 ± 7.43	66.17 ± 6.65	<0.001
Sex				0.184
Female	700 (57.5%)	535 (58.6%)	165 (54.1%)	
Male	518 (42.5%)	378 (41.4%)	140 (45.9%)	
Educational attainment				0.749
<High School	396 (32.5%)	295 (32.3%)	101 (33.1%)	
High School	467 (38.3%)	356 (39.0%)	111 (36.4%)	
University and above	350 (28.7%)	259 (28.4%)	91 (29.8%)	
Missing	5 (0.4%)	3 (0.3%)	2 (0.7%)	
Income, CNY				0.519
≤24k	51 (4.2%)	39 (4.3%)	12 (3.9%)	
24k–60k	823 (67.6%)	621 (68.0%)	202 (66.2%)	
60k–120k	169 (13.9%)	129 (14.1%)	40 (13.1%)	
≥120k	17 (1.4%)	14 (1.5%)	3 (1.0%)	
Missing	158 (13.0%)	110 (12.0%)	48 (15.7%)	
Smoking status				0.481
Ever/Current	1,074 (88.2%)	809 (88.6%)	265 (86.9%)	
Never	144 (11.8%)	104 (11.4%)	40 (13.1%)	
Alcohol				0.534
Ever/Current	1,107 (90.9%)	833 (91.2%)	274 (89.8%)	
Never	111 (9.1%)	80 (8.8%)	31 (10.2%)	
Body mass index, kg/m ²	24.53 ± 3.16	24.45 ± 3.14	24.76 ± 3.19	0.133
SBP, mmHg	133.44 ± 17.88	132.59 ± 17.47	135.97 ± 18.86	0.004
DBP, mmHg	70.77 ± 9.98	70.73 ± 9.92	70.88 ± 10.18	0.821
Duration of diabetes, year	8.39 ± 6.68	8.15 ± 6.56	9.12 ± 6.99	0.027
HbA1c, %	6.92 ± 1.29	6.94 ± 1.28	6.86 ± 1.29	0.324
Use of insulin	235 (19.3%)	177 (19.4%)	58 (19.0%)	0.950
Total cholesterol, mmol/L	4.80 ± 1.05	4.81 ± 1.06	4.78 ± 1.02	0.671
LDL-C, mmol/L	3.02 ± 0.93	3.02 ± 0.95	3.02 ± 0.89	0.982
HDL-C, mmol/L	1.30 ± 0.40	1.31 ± 0.41	1.28 ± 0.40	0.336
Serum creatine, mmol/L	70.39 ± 19.18	68.70 ± 17.61	75.45 ± 22.55	<0.001
Microalbuminuria, mg/mL	3.61 ± 10.73	2.68 ± 7.49	6.43 ± 16.85	<0.001
BCVA, LogMAR	0.18 ± 0.15	0.17 ± 0.14	0.23 ± 0.17	<0.001
Axial length, mm	23.52 ± 1.09	23.48 ± 1.07	23.64 ± 1.16	0.024
Baseline pRNFL thickness, μm	109.88 ± 12.65	109.81 ± 12.29	110.07 ± 13.67	0.756

*P values were calculated using Student *t* test for continuous variables and the χ^2 test for categorical variables. Bold values indicate statistical significance.

DR, diabetic retinopathy; SBP, systolic blood pressure; DBP, diastolic blood pressure; HbA1c, glycosylated hemoglobin; LDL-C, low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; BCVA, best-corrected visual acuity.

<https://doi.org/10.1371/journal.pmed.1004868.t001>

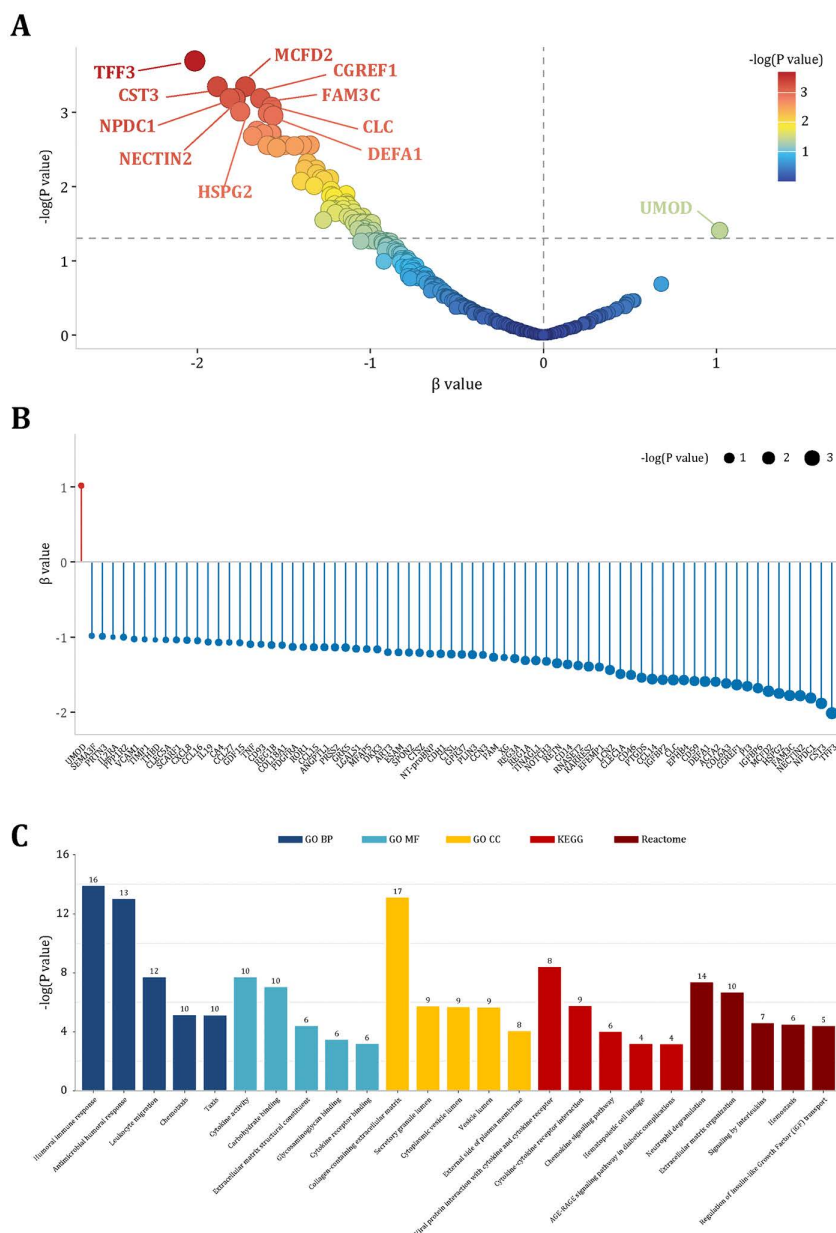


Fig 2. DRN-associated proteins and functional enrichment. (A) Volcano plot summarizing DRN-associated proteins. The x-axis represents standardized β coefficients; the y-axis shows statistical significance as $-\log_{10}(P)$ values. Highlighted proteins (e.g., TFF3, CST3, MCFD2, FAM3C, NPDC1, UMOD) exceed predefined thresholds. (B) Ranked effect sizes (lollipop plot) for all proteins; point size reflects $-\log_{10}(P)$. Most associations are negative (protective), with UMOD showing a positive effect. (C) Over-representation analysis of FDR-significant proteins across Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), and Reactome databases. Numbers above bars denote the count of proteins contributing to each term. Enriched pathways including inflammatory response, extracellular matrix remodeling, and neurotrophic signaling.

<https://doi.org/10.1371/journal.pmed.1004868.g002>

proteins most strongly linked to accelerated thinning included CST3, HSPG2, NECTIN2, COL6A3, ACTA2, NPDC1, CD59, PTGDS, RNASET2, and FAM3C, with β values ranging from $-0.445 \mu\text{m}/\text{year}$ (95% CI $[-0.518, -0.372]$) to $-0.112 \mu\text{m}/\text{year}$ (95% CI $[-0.173, -0.050]$). Notably, UMOD consistently exhibited a protective effect, with higher levels associated with thicker baseline RNFL ($\beta = 1.019 \mu\text{m}$; 95% CI $[0.273, 1.764]$) and slower thinning over time ($\beta = 0.256 \mu\text{m}/\text{year}$; 95% CI

[0.192, 0.320]). These associations between the RNFL thinning rate and proteins remained robust in sensitivity analyses ([S4 Table](#)).

Functional annotation of DRN-associated proteins

Functional enrichment analyses using the GO, Kyoto Encyclopaedia of Genes and Genomes (KEGG), and Reactome revealed four interrelated biological processes ([Fig 2C](#)): (1) inflammation and immune recruitment (Cytokine–cytokine receptor interaction, Chemokine signaling, Leukocyte migration, Neutrophil degranulation); (2) basement membrane and ECM remodeling (Collagen-containing ECM, ECM organization, Glycosaminoglycan binding, Basement membrane); (3) membrane receptor function and blood–retina barrier integrity (Cell adhesion molecules, Cytokine receptor binding); and (4) microvascular homeostasis and metabolic stress (Complement and coagulation cascades, Hemostasis, AGE–RAGE signaling). These enrichment results implicate chronic inflammation, extracellular matrix dysregulation, receptor-mediated vascular dysfunction, and microvascular-metabolic stress as interlinked molecular correlates of early DRN.

Proteomics-based DRN (Pro-DRN) prediction model

We constructed Pro-DRN using plasma protein markers identified in GDES-PPP, employing eight widely used ML algorithms: XGBoost, LightGBM, Random Forest, Neural Network, Logistic Regression, K-Nearest Neighbors, Support Vector Machine (SVM), and Decision Tree. Models were trained using an 8:2 training/test split, with hyperparameters optimized via cross-validation, and performance evaluated in an independent test set ([Fig 3](#); [S5 Table](#)).

In the training set, model C-index ranged from 0.760 to 0.984, with Random Forest 0.984 (95% CI [0.977, 0.990]), XGBoost 0.979 (95% CI [0.972, 0.986]), and SVM 0.947 (95% CI [0.933, 0.961]) performing best. In the test set, C-indices ranged from 0.736 to 0.860, with XGBoost 0.860 (95% CI [0.810, 0.911]), LightGBM 0.832 (95% CI [0.777–0.886]), and Random Forest 0.816 (95% CI [0.756–0.876]) leading. XGBoost was selected as the final model on the basis of its superior balance between discrimination, generalisability, and training stability, and was used for all subsequent Pro-DRN scoring, feature-importance interpretation, and clinical utility analyses.

Importance ranking of Pro-DRN plasma proteins

To interpret the model, we applied Shapley values, providing both global and individual-level insights while accounting for protein collinearity and complex interactions ([Fig 4](#)). Globally, the top 10 proteins contributing to DRN risk were ACTA2, COL6A3, CCL27, NT-proBNP, CD46, MCFD2, CTSZ, CCL15, RARRES2, and HSPG2 ([Fig 4A](#)). Beeswarm plots confirmed consistent effect directions across individual samples, with higher expression of most top proteins increasing predicted risk.

In high-risk individuals (predicted probability=0.85), COL6A3, ACTA2, CCL16, REG3A, HSPG2, PAM, ART3, and TIMP1 jointly drove risk, demonstrating a multi-protein additive effect ([Fig 4B](#)). The concordance between global and individual explanations indicates that the discriminative power of Pro-DRN arises from the integrated contributions of multiple proteins rather than from any single dominant feature, a property that simultaneously enhances interpretability and predictive robustness.

Pro-DRN substantially improves DRN risk discrimination

We systematically evaluated the predictive performance of Pro-DRN in the test set. As a standalone predictor, Pro-DRN achieved excellent discrimination (C-index=0.860; 95% CI [0.810, 0.911]), far exceeding any single conventional predictor (C-index: 0.500 to 0.688; [S6 Table](#)), highlighting the independent and superior value of plasma proteomic features. Integrating Pro-DRN with a baseline model of age and sex increased the C-index from 0.697 to 0.877 (95% CI [0.834, 0.921]), representing a 25.9% relative improvement ([Table 2](#)).

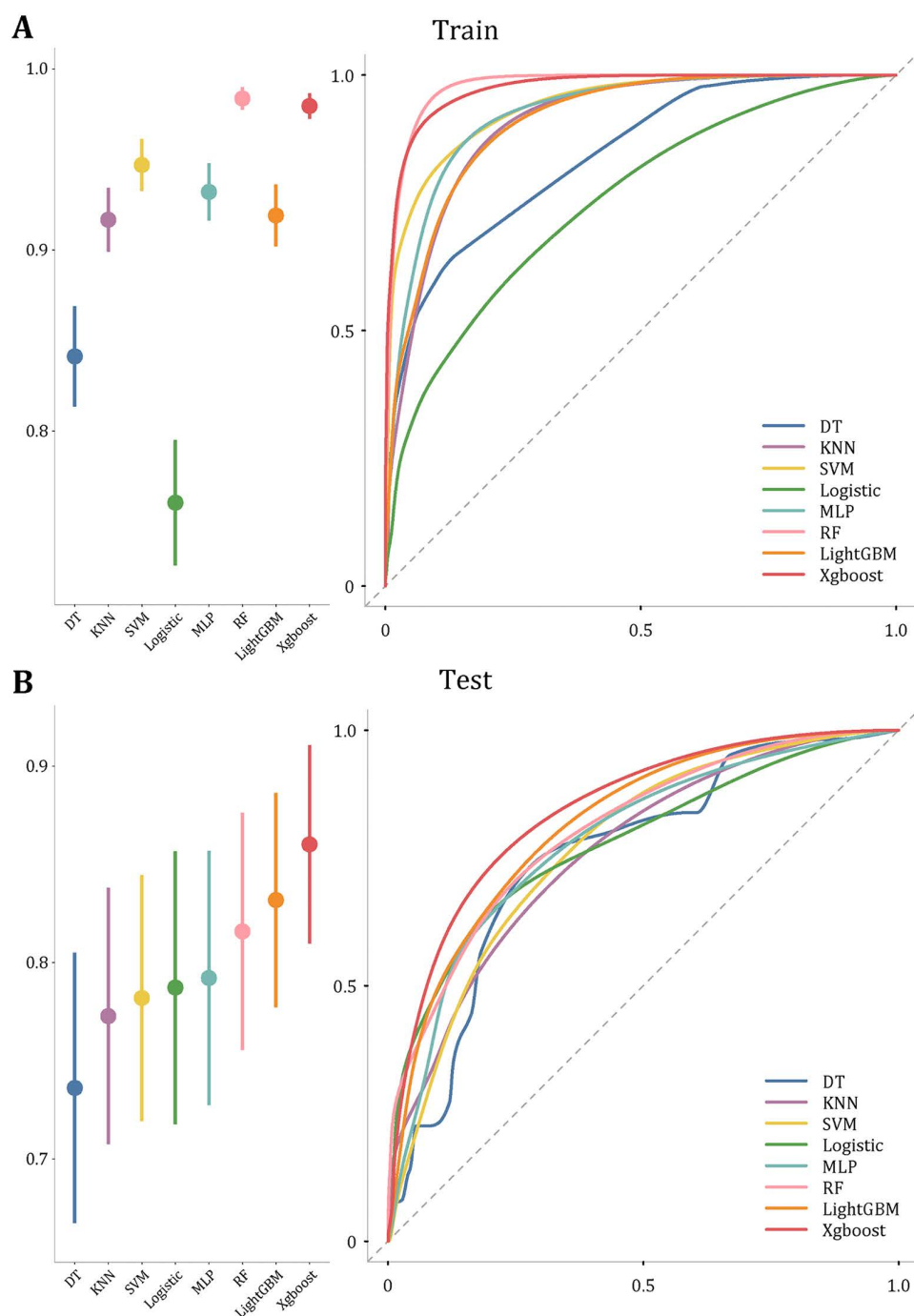


Fig 3. Performance of machine-learning models for DRN prediction. (A) Training set. Left, model discrimination summarized as AUC (C-index) with 95% confidence intervals; right, ROC curves for each model. **(B)** Held-out test set, displayed as in **(A)**. The dashed diagonal denotes chance performance (AUC=0.5). Across both sets, gradient-boosting models (XGBoost and LightGBM) show the highest discrimination. **Abbreviations:** DT, decision tree; KNN, k-nearest neighbors; SVM, support vector machine; MLP, multilayer perceptron; RF, random forest.

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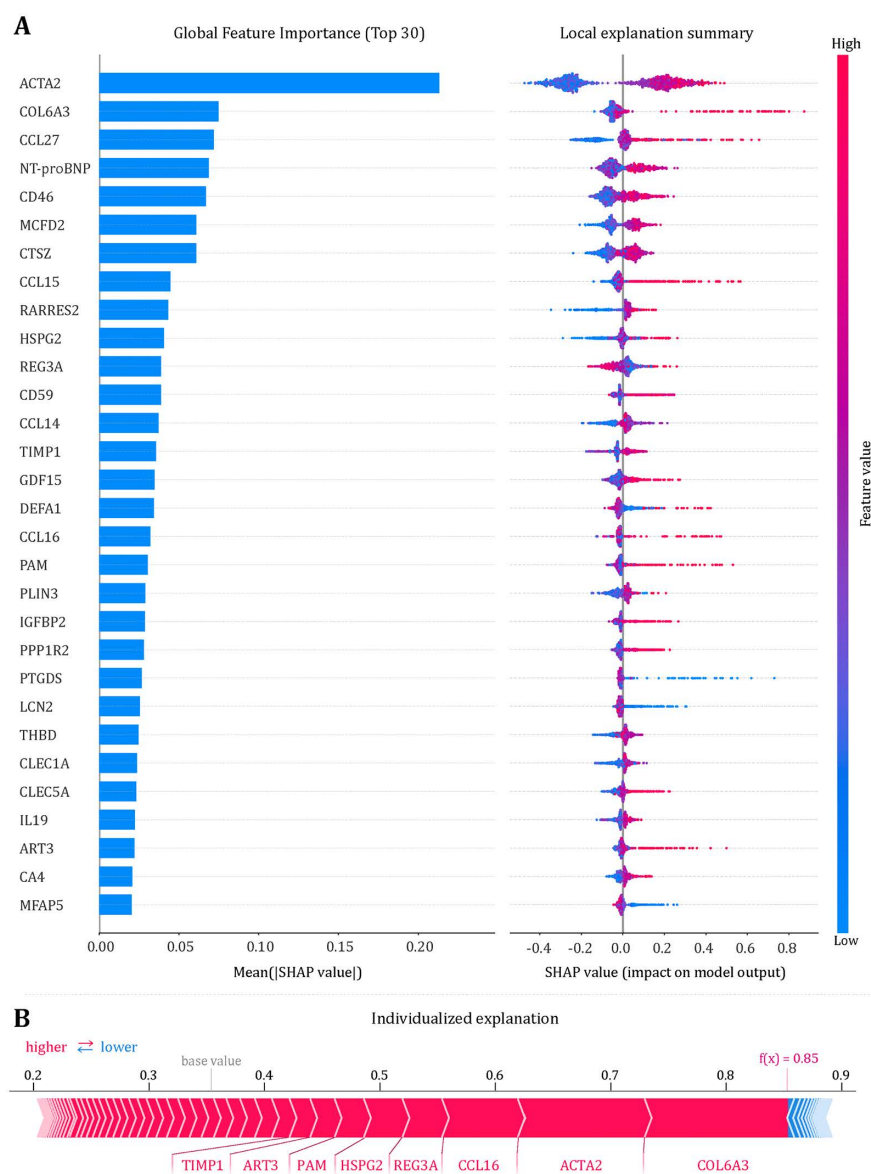


Fig 4. Global and individual explanations of the Pro-DRN model. (A) XGBoost interpretation. Left: global feature importance ranked by mean [SHAP] for the top 30 proteins (longer bars = greater overall contribution). Right: SHAP beeswarm plot showing the signed effect of each feature on the predicted probability of DRN (x-axis, SHAP value); red/blue indicate higher/lower feature values. High-impact proteins include ACTA2, COL6A3, CCL27, NT-proBNP, CD46, MCFD2, CTSZ, CCL15, RARRES2 and HSPG2. **(B)** Individual explanation (force plot) for a representative participant with predicted risk $f(x) = 0.85$. Red features push the prediction above the model base value (increase risk), and blue features pull it down; arrow width reflects absolute SHAP contribution.

<https://doi.org/10.1371/journal.pmed.1004868.g004>

To assess incremental values, Pro-DRN was added to six established clinical models (Fig 5A and 5B). All models showed consistent gains in discrimination, with average Δ C-index ranging from 0.137 to 0.159, and the fully adjusted model (all clinical variables plus Pro-DRN) achieved a C-index of 0.908 (95% CI [0.869, 0.948]; Table 2). Importantly, the Pro-DRN model achieved discrimination that was comparable to, and modestly higher than, the model built using all measured proteins (S7 Table). Risk reclassification analyses further confirmed its value: average IDI increased by 23.3%

Table 2. Incremental predictive value of the Pro-DRN across established models.

	C-index (95% CI)		Improvement	P value*
	Benchmark			Incorporating Pro-DRN
XGboost model				
Age&Sex	0.697 (0.626, 0.768)	0.877 (0.834, 0.921)	25.87%	4.20 × 10 ⁻⁰⁷
Aspelund model	0.733 (0.665, 0.800)	0.888 (0.847, 0.930)	21.28%	3.07 × 10 ⁻⁰⁶
Hippisley model	0.739 (0.670, 0.808)	0.898 (0.858, 0.937)	21.52%	2.31 × 10 ⁻⁰⁶
Dagliati model	0.743 (0.676, 0.809)	0.898 (0.859, 0.938)	20.96%	1.12 × 10 ⁻⁰⁶
ISDR model	0.733 (0.665, 0.801)	0.889 (0.847, 0.930)	21.20%	3.80 × 10 ⁻⁰⁶
JDC model	0.748 (0.681, 0.815)	0.889 (0.848, 0.930)	18.85%	8.91 × 10 ⁻⁰⁶
Tarasewicz model	0.760 (0.694, 0.827)	0.897 (0.857, 0.937)	18.01%	2.16 × 10 ⁻⁰⁵
All model	0.812 (0.750, 0.874)	0.908 (0.869, 0.948)	11.89%	5.19 × 10 ⁻⁰⁴
LightGBM model				
Age&Sex	0.697 (0.626, 0.768)	0.848 (0.799, 0.897)	21.67%	1.32 × 10 ⁻⁰⁵
Aspelund model	0.733 (0.665, 0.800)	0.860 (0.814, 0.906)	17.38%	1.03 × 10 ⁻⁰⁴
Hippisley model	0.739 (0.670, 0.808)	0.875 (0.830, 0.919)	18.36%	4.73 × 10 ⁻⁰⁵
Dagliati model	0.743 (0.676, 0.809)	0.873 (0.829, 0.917)	17.57%	2.85 × 10 ⁻⁰⁵
ISDR model	0.733 (0.665, 0.801)	0.861 (0.815, 0.907)	17.49%	1.05 × 10 ⁻⁰⁴
JDC model	0.748 (0.681, 0.815)	0.862 (0.816, 0.908)	15.24%	2.25 × 10 ⁻⁰⁴
Tarasewicz model	0.760 (0.694, 0.827)	0.873 (0.828, 0.917)	14.79%	2.61 × 10 ⁻⁰⁴
All model	0.812 (0.750, 0.874)	0.891 (0.850, 0.932)	9.74%	0.002
RF model				
Age&Sex	0.697 (0.626, 0.768)	0.844 (0.791, 0.897)	21.06%	4.06 × 10 ⁻⁰⁵
Aspelund model	0.733 (0.665, 0.800)	0.868 (0.822, 0.915)	18.51%	8.37 × 10 ⁻⁰⁵
Hippisley model	0.739 (0.670, 0.808)	0.872 (0.824, 0.919)	18.02%	8.21 × 10 ⁻⁰⁵
Dagliati model	0.743 (0.676, 0.809)	0.880 (0.837, 0.924)	18.55%	2.10 × 10 ⁻⁰⁵
ISDR model	0.733 (0.665, 0.801)	0.868 (0.822, 0.915)	18.45%	1.01 × 10 ⁻⁰⁴
JDC model	0.748 (0.681, 0.815)	0.868 (0.822, 0.915)	16.08%	2.40 × 10 ⁻⁰⁴
Tarasewicz model	0.760 (0.694, 0.827)	0.875 (0.828, 0.921)	15.03%	4.09 × 10 ⁻⁰⁴
All model	0.812 (0.750, 0.874)	0.897 (0.856, 0.937)	10.47%	0.002
NN model				
Age&Sex	0.697 (0.626, 0.768)	0.814 (0.757, 0.871)	16.75%	6.99 × 10 ⁻⁰⁴
Aspelund model	0.733 (0.665, 0.800)	0.835 (0.784, 0.886)	14.00%	0.001
Hippisley model	0.739 (0.670, 0.808)	0.842 (0.790, 0.895)	14.00%	6.52 × 10 ⁻⁰⁴
Dagliati model	0.743 (0.676, 0.809)	0.848 (0.799, 0.898)	14.26%	5.29 × 10 ⁻⁰⁴
ISDR model	0.733 (0.665, 0.801)	0.839 (0.788, 0.889)	14.41%	0.001
JDC model	0.748 (0.681, 0.815)	0.842 (0.791, 0.893)	12.58%	0.001
Tarasewicz model	0.760 (0.694, 0.827)	0.854 (0.804, 0.903)	12.28%	0.001
All model	0.812 (0.750, 0.874)	0.871 (0.824, 0.917)	7.23%	0.010
Logistic model				
Age&Sex	0.697 (0.626, 0.768)	0.824 (0.767, 0.881)	18.21%	3.51 × 10 ⁻⁰⁴
Aspelund model	0.733 (0.665, 0.800)	0.841 (0.788, 0.894)	14.82%	8.63 × 10 ⁻⁰⁴
Hippisley model	0.739 (0.670, 0.808)	0.849 (0.795, 0.902)	14.86%	2.74 × 10 ⁻⁰⁴
Dagliati model	0.743 (0.676, 0.809)	0.854 (0.803, 0.905)	15.00%	2.16 × 10 ⁻⁰⁴
ISDR model	0.733 (0.665, 0.801)	0.844 (0.792, 0.897)	15.17%	8.07 × 10 ⁻⁰⁴
JDC model	0.748 (0.681, 0.815)	0.846 (0.793, 0.899)	13.12%	0.001
Tarasewicz model	0.760 (0.694, 0.827)	0.860 (0.810, 0.909)	13.05%	7.99 × 10 ⁻⁰⁴

(Continued)

Table 2. (Continued)

	C-index (95% CI)		Improvement	P value*
	Benchmark			
All model	0.812 (0.750, 0.874)	0.875 (0.825, 0.924)	7.76%	0.007
KNN model				
Age&Sex	0.697 (0.626, 0.768)	0.816 (0.760, 0.873)	17.12%	3.74×10^{-04}
Aspelund model	0.733 (0.665, 0.800)	0.841 (0.790, 0.891)	14.75%	6.34×10^{-04}
Hippisley model	0.739 (0.670, 0.808)	0.852 (0.802, 0.901)	15.27%	2.53×10^{-04}
Dagliati model	0.743 (0.676, 0.809)	0.851 (0.803, 0.899)	14.57%	3.31×10^{-04}
ISDR model	0.733 (0.665, 0.801)	0.839 (0.789, 0.890)	14.49%	8.01×10^{-04}
JDC model	0.748 (0.681, 0.815)	0.842 (0.792, 0.892)	12.60%	0.002
Tarasewicz model	0.760 (0.694, 0.827)	0.852 (0.804, 0.901)	12.10%	0.001
All model	0.812 (0.750, 0.874)	0.875 (0.830, 0.919)	7.74%	0.008
SVM model				
Age&Sex	0.697 (0.626, 0.768)	0.810 (0.755, 0.866)	16.26%	6.73×10^{-04}
Aspelund model	0.733 (0.665, 0.800)	0.826 (0.774, 0.878)	12.73%	0.002
Hippisley model	0.739 (0.670, 0.808)	0.838 (0.786, 0.890)	13.43%	7.55×10^{-04}
Dagliati model	0.743 (0.676, 0.809)	0.834 (0.783, 0.886)	12.36%	0.001
ISDR model	0.733 (0.665, 0.801)	0.828 (0.776, 0.879)	12.92%	0.002
JDC model	0.748 (0.681, 0.815)	0.826 (0.773, 0.879)	10.49%	0.003
Tarasewicz model	0.760 (0.694, 0.827)	0.841 (0.791, 0.892)	10.65%	0.003
All model	0.812 (0.750, 0.874)	0.865 (0.816, 0.914)	6.55%	0.010
DT model				
Age&Sex	0.697 (0.626, 0.768)	0.760 (0.697, 0.824)	9.09%	0.012
Aspelund model	0.733 (0.665, 0.800)	0.795 (0.737, 0.853)	8.52%	0.010
Hippisley model	0.739 (0.670, 0.808)	0.793 (0.734, 0.852)	7.33%	0.011
Dagliati model	0.743 (0.676, 0.809)	0.803 (0.747, 0.859)	8.13%	0.008
ISDR model	0.733 (0.665, 0.801)	0.796 (0.739, 0.853)	8.53%	0.009
JDC model	0.748 (0.681, 0.815)	0.805 (0.746, 0.864)	7.64%	0.009
Tarasewicz model	0.760 (0.694, 0.827)	0.820 (0.765, 0.876)	7.89%	0.005
All model	0.812 (0.750, 0.874)	0.845 (0.793, 0.898)	4.13%	0.039

*P values were calculated using DeLong's test for paired comparisons of C-indices between the benchmark model and the corresponding model incorporating Pro-DRN.

Pro-DRN, proteome-based diabetic retinal neurodegeneration; CI, confidence interval; RF, random forest; NN, neural network; KNN, K-Nearest Neighbors; SVM, support vector machine; DT, decision tree.

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(range 21.2% to 24.5%) and NRI by 38.7% (range 22.6% to 45.2%), with the All model showing IDI = 18.3% (95% CI [0.120, 0.248]; $P < 0.001$) and NRI = 35.5% (95% CI [0.119, 0.593]; $P = 0.008$; [Table 3](#)).

To evaluate robustness across algorithms, Pro-DRN was reconstructed using seven alternative ML methods. All versions consistently improved discrimination and reclassification. For instance, LightGBM achieved Δ C-index values ranging from 0.113 to 0.136 (best C-index = 0.891; 95% CI [0.850, 0.932]), with IDI = 17.9% and NRI = 38.7%. Random Forest yielded Δ C-index values ranging from 0.115 to 0.137 (best C-index = 0.897; 95% CI [0.856–0.937]), with IDI = 21.6% and NRI = 29.6% ([Tables 2](#) and [3](#)). These consistent improvements across diverse algorithms underscore the reproducibility, generalizability, and robustness of proteomic markers for enhancing DRN risk stratification.

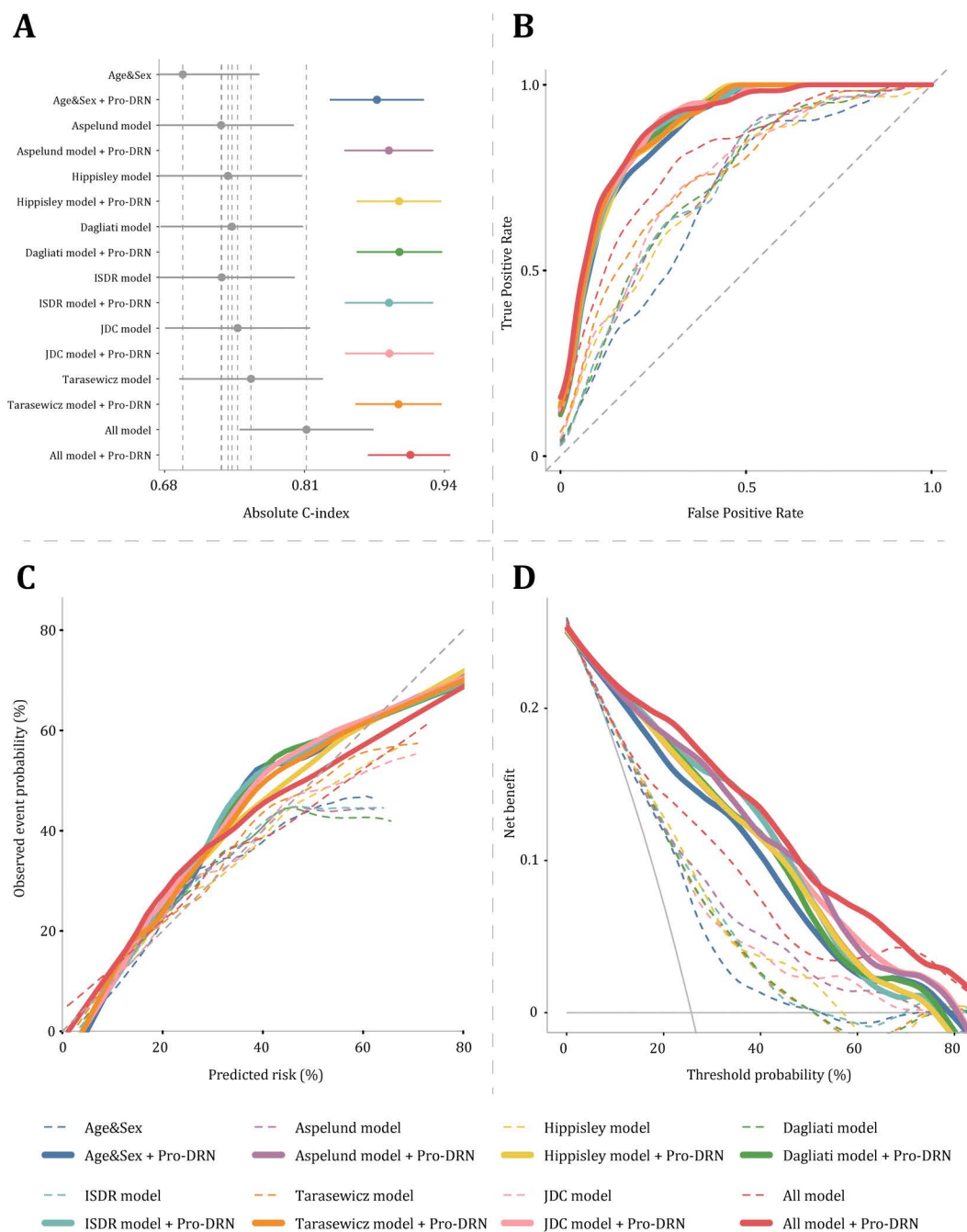


Fig 5. Pro-DRN enhances conventional DRN prediction models. (A) Discrimination for eight clinical frameworks with and without the proteomics score (Pro-DRN). Gray points and horizontal bars denote the original models with 95% CIs; colored points indicate models after adding Pro-DRN. Across frameworks, Pro-DRN consistently improves the C-index, with the “All model + Pro-DRN” performing best. (B) ROC curves. Solid lines show models with Pro-DRN and dashed lines the corresponding clinical models. Pro-DRN shifts the curves upward and left. (C) Calibration (observed event probability vs. predicted risk). Models augmented with Pro-DRN (solid) align more closely with the 45° reference line than their baselines (dashed). (D) Decision curve analysis. Net benefit across threshold probabilities is higher after adding Pro-DRN over a broad range of thresholds.

<https://doi.org/10.1371/journal.pmed.1004868.g005>

Table 3. Reclassification and discrimination gains with Pro-DRN across established models.

	IDI (95% CI)	P value*	NRI (95% CI)	P value*
XGboost model				
Age&Sex	0.242 (0.170, 0.316)	<0.001	0.419 (0.180, 0.643)	<0.001
Aspelund model	0.241 (0.170, 0.314)	<0.001	0.452 (0.222, 0.673)	<0.001
Hippisley model	0.245 (0.177, 0.318)	<0.001	0.452 (0.222, 0.673)	<0.001
Dagliati model	0.242 (0.175, 0.315)	<0.001	0.355 (0.123, 0.593)	0.004
ISDR model	0.240 (0.170, 0.312)	<0.001	0.452 (0.222, 0.673)	<0.001
JDC model	0.212 (0.145, 0.285)	<0.001	0.226 (0.030, 0.467)	0.028
Tarasewicz model	0.216 (0.147, 0.291)	<0.001	0.387 (0.147, 0.607)	0.002
All model	0.183 (0.120, 0.248)	<0.001	0.355 (0.119, 0.593)	0.008
LightGBM model				
Age&Sex	0.195 (0.130, 0.258)	<0.001	0.290 (0.040, 0.524)	0.030
Aspelund model	0.185 (0.119, 0.245)	<0.001	0.387 (0.129, 0.618)	0.002
Hippisley model	0.193 (0.123, 0.254)	<0.001	0.452 (0.200, 0.672)	<0.001
Dagliati model	0.193 (0.128, 0.253)	<0.001	0.484 (0.250, 0.683)	<0.001
ISDR model	0.186 (0.120, 0.246)	<0.001	0.387 (0.129, 0.618)	0.002
JDC model	0.156 (0.096, 0.211)	<0.001	0.323 (0.085, 0.541)	0.008
Tarasewicz model	0.165 (0.109, 0.230)	<0.001	0.290 (0.064, 0.539)	0.018
All model	0.135 (0.087, 0.190)	<0.001	0.419 (0.193, 0.652)	0.002
RF model				
Age&Sex	0.222 (0.150, 0.292)	<0.001	0.323 (0.079, 0.545)	0.014
Aspelund model	0.224 (0.151, 0.293)	<0.001	0.290 (0.059, 0.514)	0.022
Hippisley model	0.227 (0.154, 0.293)	<0.001	0.258 (0.015, 0.491)	0.048
Dagliati model	0.228 (0.157, 0.297)	<0.001	0.323 (0.079, 0.545)	0.014
ISDR model	0.224 (0.150, 0.293)	<0.001	0.290 (0.059, 0.514)	0.022
JDC model	0.195 (0.125, 0.260)	<0.001	0.323 (0.068, 0.548)	0.016
Tarasewicz model	0.198 (0.133, 0.273)	<0.001	0.290 (0.051, 0.539)	0.018
All model	0.166 (0.105, 0.232)	<0.001	0.258 (0.014, 0.500)	0.046
NN model				
Age&Sex	0.143 (0.086, 0.196)	<0.001	0.258 (0.000, 0.491)	0.052
Aspelund model	0.136 (0.080, 0.189)	<0.001	0.226 (−0.030, 0.469)	0.090
Hippisley model	0.142 (0.087, 0.196)	<0.001	0.226 (−0.030, 0.469)	0.090
Dagliati model	0.144 (0.089, 0.199)	<0.001	0.290 (0.033, 0.525)	0.036
ISDR model	0.143 (0.084, 0.197)	<0.001	0.290 (0.019, 0.527)	0.034
JDC model	0.115 (0.061, 0.166)	<0.001	0.323 (0.071, 0.556)	0.016
Tarasewicz model	0.121 (0.070, 0.177)	<0.001	0.258 (0.017, 0.524)	0.032
All model	0.092 (0.047, 0.144)	<0.001	0.226 (0.000, 0.491)	0.074
Logistic model				
Age&Sex	0.182 (0.120, 0.249)	<0.001	0.323 (0.094, 0.541)	<0.001
Aspelund model	0.178 (0.118, 0.244)	<0.001	0.355 (0.125, 0.568)	<0.001
Hippisley model	0.179 (0.119, 0.245)	<0.001	0.290 (0.053, 0.508)	0.014
Dagliati model	0.184 (0.123, 0.251)	<0.001	0.355 (0.129, 0.574)	<0.001
ISDR model	0.184 (0.122, 0.250)	<0.001	0.355 (0.123, 0.571)	<0.001
JDC model	0.154 (0.096, 0.212)	<0.001	0.258 (0.016, 0.483)	0.040
Tarasewicz model	0.154 (0.099, 0.215)	<0.001	0.258 (0.032, 0.510)	0.036
All model	0.129 (0.076, 0.189)	<0.001	0.290 (0.069, 0.531)	0.012

(Continued)

Table 3. (Continued)

	IDI (95% CI)	P value*	NRI (95% CI)	P value*
KNN model				
Age&Sex	0.140 (0.084, 0.200)	<0.001	0.161 (−0.100, 0.417)	0.234
Aspelund model	0.133 (0.078, 0.190)	<0.001	0.161 (−0.100, 0.412)	0.240
Hippisley model	0.144 (0.087, 0.200)	<0.001	0.194 (−0.063, 0.444)	0.152
Dagliati model	0.133 (0.079, 0.189)	<0.001	0.161 (−0.091, 0.404)	0.230
ISDR model	0.134 (0.078, 0.191)	<0.001	0.194 (−0.069, 0.440)	0.182
JDC model	0.107 (0.057, 0.158)	<0.001	0.161 (−0.085, 0.397)	0.200
Tarasewicz model	0.105 (0.057, 0.156)	<0.001	0.161 (−0.063, 0.433)	0.210
All model	0.087 (0.042, 0.137)	<0.001	0.226 (0.000, 0.490)	0.068
SVM model				
Age&Sex	0.124 (0.073, 0.174)	<0.001	0.161 (−0.075, 0.393)	0.242
Aspelund model	0.118 (0.068, 0.167)	<0.001	0.194 (−0.057, 0.429)	0.150
Hippisley model	0.125 (0.073, 0.176)	<0.001	0.258 (0.014, 0.482)	0.046
Dagliati model	0.123 (0.071, 0.176)	<0.001	0.226 (−0.029, 0.458)	0.084
ISDR model	0.119 (0.069, 0.169)	<0.001	0.129 (−0.108, 0.367)	0.378
JDC model	0.092 (0.049, 0.135)	<0.001	0.226 (−0.016, 0.465)	0.084
Tarasewicz model	0.092 (0.047, 0.137)	<0.001	0.161 (−0.071, 0.410)	0.196
All model	0.073 (0.036, 0.114)	0.002	0.258 (0.019, 0.500)	0.038
DT model				
Age&Sex	0.064 (0.026, 0.106)	<0.001	−0.097 (−0.334, 0.161)	0.480
Aspelund model	0.061 (0.024, 0.102)	0.002	−0.097 (−0.334, 0.161)	0.480
Hippisley model	0.062 (0.025, 0.104)	0.004	−0.065 (−0.300, 0.194)	0.656
Dagliati model	0.065 (0.028, 0.105)	<0.001	−0.097 (−0.334, 0.161)	0.480
ISDR model	0.061 (0.024, 0.102)	0.004	−0.097 (−0.334, 0.161)	0.480
JDC model	0.054 (0.020, 0.092)	0.002	−0.065 (−0.300, 0.208)	0.654
Tarasewicz model	0.056 (0.017, 0.095)	0.002	−0.065 (−0.310, 0.192)	0.646
All model	0.050 (0.013, 0.090)	0.012	0.000 (−0.236, 0.262)	1.046

*P values for IDI and NRI were estimated by bootstrap resampling.

Pro-DRN, proteome-based diabetic retinal neurodegeneration; IDI, integrated discrimination improvement; NRI, net reclassification improvement; CI, Confidence interval; RF, random forest; NN, neural network; KNN, K-Nearest Neighbors; SVM, support vector machine; DT, decision tree.

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Pro-DRN demonstrates excellent calibration and clinical utility

Beyond discrimination and reclassification, we systematically evaluated Pro-DRN's calibration and clinical utility (Fig 5C and 5D). Calibration plots showed that all conventional models incorporating Pro-DRN achieved markedly improved agreement between predicted and observed DRN risk, with curves closely approaching the ideal diagonal and substantially reducing systematic bias, particularly in mid-to-high risk ranges (Fig 5C). Decision curve analysis (DCA) further demonstrated that models with Pro-DRN consistently delivered higher net clinical benefit across relevant threshold probabilities, outperforming both the original clinical models and extreme strategies of “treat all” or “treat none.” Notably, the fully adjusted model combined with Pro-DRN achieved the greatest net benefit across most thresholds (Fig 5D). These results indicate that Pro-DRN substantially enhances calibration and clinical utility, offering a robust, interpretable, and actionable tool for early identification and targeted intervention in individuals at high risk of DRN.

Robustness Analyses of Pro-DRN Integrated Models

To further stress-test the benchmarks, we additionally augmented each conventional model with measures of insulin resistance (fasting insulin, HOMA2-IR, TyG), beta cell function (fasting C-peptide, HOMA2-B), and glycemic indices (fasting glucose, HbA1c); Pro-DRN remained consistently additive across all augmented baselines (S8 Table).

We tested robustness using three alternative DRN definitions (DRN-Top10Slope, DRN-ThinLast and DRN-ExcessLoss). Pro-DRN preserved discrimination across all definitions (C-index 0.880, 0.753 and 0.848, respectively; S9 Table) and consistently improved the fully adjusted clinical framework (All model: 0.856 to 0.935; 0.738 to 0.832; and 0.823 to 0.902; S10 Table).

To establish the clinical relevance of Pro-DRN beyond its OCT-based definition, we examined whether the baseline Pro-DRN score was associated with subsequent patient-centered outcomes. Higher Pro-DRN was associated with subsequent clinically meaningful vision impairment defined by 15 or more ETDRS letters loss (OR = 27.98; 95% CI [1.07, 578.13]; $P=0.036$), and was also associated with diabetic peripheral neuropathy (OR = 5.82; 95% CI [1.41, 22.97]; $P=0.013$) as defined by the Michigan Neuropathy Screening Instrument patient questionnaire (Table 4). These findings support the broader clinical relevance of Pro-DRN, although they should be interpreted cautiously given the limited number of events and the wide confidence interval.

Shiny-based online tool for clinical translation

To accelerate clinical translation, we deployed the locked Pro-DRN model as an interactive, web-based decision-support tool (https://fmb-gdes2025.shinyapps.io/DRN_predict/). The tool comprises two core modules: (1) Pro-DRN score calculation: users can upload standardized measurements of DRN-associated plasma proteins, and the system computes the Pro-DRN score in real time using the XGBoost model developed in this study; (2) DRN risk prediction: based on the calculated score, the tool generates individualized risk estimates with dynamic visual feedback. The interface supports real-time parameter linkage, input validation, and continuous risk updating as data change. An intuitive risk-stratification display further aids clinical interpretation. Designed with a “left input–right output” layout, the platform enables rapid, reliable risk assessment and documentation, making it suitable for clinical consultations, screening programs, and longitudinal follow-up.

Cross-ethnic external validation in UKB-PPP

To assess generalizability and robustness, we performed cross-ethnic external validation in the independent cohort. Despite limited longitudinal proteome–imaging data and platform differences, we systematically examined cross-sectional associations between plasma proteins and macular RNFL thickness using standardized covariate adjustment and FDR correction (Table 5). In univariate analyses, 52 of 71 proteins (73.2%) were significantly negatively associated with RNFL thickness, with β values ranging from $-0.841\ \mu\text{m}$ (95% CI $[-1.206, -0.476]$) to $-0.337\ \mu\text{m}$ (95% CI $[-0.618, -0.056]$) per 1–SD increase. UMOD was the sole protein with a positive association ($\beta=0.488$; 95% CI $[0.171, 0.805]$). In multi-variable models adjusting for confounders and multiple testing, 34 proteins retained significant negative associations,

Table 4. Association of the Pro-DRN score with vision impairment and diabetic peripheral neuropathy.

Outcome	OR (95% CI)	P value*
Vision impairment	27.98 (1.07, 578.13)	0.036
Diabetic peripheral neuropathy	5.82 (1.41, 22.97)	0.013

*P values were obtained from logistic regression models with Wald tests.

OR, odds ratios; CI, confidence intervals.

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Table 5. Associations of proteins with retinal nerve fiber layer thickness in UK Biobank.

Proteins*	Univariable Regression			Multivariable Regression		
	β (95% CI) †	P value	P _{FDR} value‡	β (95% CI)†	P value	P _{FDR} value‡
DEFA1	−0.723 (−1.062, −0.383)	3.59 × 10^{−05}	2.35 × 10^{−04}	−0.668 (−1.014, −0.322)	1.73 × 10^{−04}	0.004
CD14	−0.586 (−0.929, −0.244)	8.67 × 10^{−04}	0.002	−0.716 (−1.077, −0.355)	1.16 × 10^{−04}	0.004
ESAM	−0.823 (−1.141, −0.504)	5.79 × 10^{−07}	4.17 × 10^{−05}	−0.696 (−1.036, −0.356)	6.89 × 10^{−05}	0.004
EPHB4	−0.762 (−1.096, −0.428)	9.54 × 10^{−06}	1.47 × 10^{−04}	−0.639 (−0.983, −0.294)	3.09 × 10^{−04}	0.006
CD93	−0.729 (−1.088, −0.370)	7.96 × 10^{−05}	3.62 × 10^{−04}	−0.671 (−1.041, −0.301)	4.22 × 10^{−04}	0.006
PI3	−0.660 (−0.974, −0.345)	4.63 × 10^{−05}	2.78 × 10^{−04}	−0.556 (−0.883, −0.228)	0.001	0.008
CD46	−0.679 (−1.020, −0.337)	1.11 × 10^{−04}	3.81 × 10^{−04}	−0.590 (−0.937, −0.243)	0.001	0.008
RNASET2	−0.703 (−1.019, −0.387)	1.63 × 10^{−05}	1.47 × 10^{−04}	−0.558 (−0.889, −0.227)	0.001	0.008
EFEMP1	−0.687 (−1.017, −0.356)	5.43 × 10^{−05}	2.79 × 10^{−04}	−0.589 (−0.943, −0.236)	0.001	0.008
CCL15	−0.599 (−0.903, −0.294)	1.32 × 10^{−04}	4.14 × 10^{−04}	−0.525 (−0.838, −0.211)	0.001	0.008
CCN3	−0.750 (−1.085, −0.414)	1.46 × 10^{−05}	1.47 × 10^{−04}	−0.619 (−0.974, −0.265)	0.001	0.008
NPDC1	−0.682 (−1.021, −0.343)	9.15 × 10^{−05}	3.81 × 10^{−04}	−0.578 (−0.934, −0.222)	0.002	0.009
TIMP1	−0.714 (−1.032, −0.396)	1.34 × 10^{−05}	1.47 × 10^{−04}	−0.543 (−0.878, −0.208)	0.002	0.009
CCL16	−0.679 (−0.993, −0.364)	2.83 × 10^{−05}	2.04 × 10^{−04}	−0.524 (−0.855, −0.193)	0.002	0.010
GDF15	−0.724 (−1.023, −0.426)	2.60 × 10^{−06}	9.35 × 10^{−05}	−0.568 (−0.926, −0.209)	0.002	0.010
COL6A3	−0.679 (−1.013, −0.344)	8.04 × 10^{−05}	3.62 × 10^{−04}	−0.522 (−0.865, −0.178)	0.003	0.013
CD59	−0.795 (−1.145, −0.444)	1.12 × 10^{−05}	1.47 × 10^{−04}	−0.571 (−0.947, −0.196)	0.003	0.013
ACTA2	−0.841 (−1.206, −0.476)	7.99 × 10^{−06}	1.47 × 10^{−04}	−0.619 (−1.034, −0.204)	0.004	0.013
SPON2	−0.630 (−0.982, −0.277)	5.07 × 10^{−04}	0.001	−0.534 (−0.889, −0.179)	0.003	0.013
NTproBNP	−0.589 (−0.901, −0.278)	2.31 × 10^{−04}	0.001	−0.498 (−0.832, −0.165)	0.004	0.013
NECTIN2	−0.697 (−1.050, −0.345)	1.20 × 10^{−04}	3.92 × 10^{−04}	−0.527 (−0.893, −0.160)	0.005	0.015
HSPG2	−0.630 (−0.947, −0.314)	1.10 × 10^{−04}	3.81 × 10^{−04}	−0.479 (−0.808, −0.150)	0.004	0.015
PTGDS	−0.754 (−1.099, −0.410)	2.18 × 10^{−05}	1.74 × 10^{−04}	−0.536 (−0.908, −0.164)	0.005	0.015
COL18A1	−0.661 (−0.992, −0.329)	1.07 × 10^{−04}	3.81 × 10^{−04}	−0.514 (−0.870, −0.159)	0.005	0.015
GPR37	−0.636 (−0.962, −0.311)	1.47 × 10^{−04}	4.40 × 10^{−04}	−0.462 (−0.804, −0.121)	0.008	0.023
LGALS1	−0.605 (−0.959, −0.251)	8.64 × 10^{−04}	0.002	−0.486 (−0.843, −0.128)	0.008	0.023
PAM	−0.463 (−0.809, −0.116)	0.009	0.016	−0.466 (−0.814, −0.119)	0.009	0.023
TFF3	−0.808 (−1.275, −0.341)	7.54 × 10^{−04}	0.002	−0.658 (−1.150, −0.166)	0.009	0.023
CST3	−0.697 (−1.032, −0.362)	5.37 × 10^{−05}	2.79 × 10^{−04}	−0.482 (−0.844, −0.119)	0.010	0.024
LCN2	−0.506 (−0.829, −0.183)	0.002	0.004	−0.445 (−0.791, −0.099)	0.012	0.029
PLIN3	−0.579 (−0.947, −0.210)	0.002	0.004	−0.479 (−0.852, −0.105)	0.012	0.029
CGREF1	−0.527 (−0.840, −0.214)	0.001	0.002	−0.399 (−0.717, −0.082)	0.014	0.031
CDH1	−0.613 (−0.995, −0.232)	0.002	0.004	−0.485 (−0.872, −0.098)	0.014	0.031
VCAM1	−0.513 (−0.830, −0.196)	0.002	0.004	−0.417 (−0.750, −0.084)	0.015	0.031
SEMA3F	−0.515 (−0.842, −0.188)	0.002	0.004	−0.385 (−0.721, −0.048)	0.026	0.053
TINAGL1	−0.427 (−0.776, −0.078)	0.017	0.025	−0.405 (−0.765, −0.044)	0.028	0.055
TNF	−0.479 (−0.803, −0.154)	0.004	0.007	−0.371 (−0.700, −0.041)	0.028	0.055
MCFD2	−0.428 (−0.755, −0.101)	0.011	0.017	−0.356 (−0.685, −0.027)	0.034	0.063
NOTCH3	−0.268 (−0.586, 0.051)	0.100	0.120	−0.372 (−0.716, −0.027)	0.035	0.063
SCARF1	−0.407 (−0.741, −0.072)	0.018	0.026	−0.368 (−0.706, −0.030)	0.034	0.063
PPP1R2	−0.448 (−0.806, −0.090)	0.014	0.022	−0.376 (−0.736, −0.016)	0.041	0.072
IGFBP6	−0.665 (−0.998, −0.331)	1.07 × 10^{−04}	3.81 × 10^{−04}	−0.408 (−0.801, −0.015)	0.042	0.072
CCL14	−0.376 (−0.704, −0.048)	0.025	0.035	−0.338 (−0.668, −0.009)	0.045	0.075
ANGPTL1	−0.259 (−0.586, 0.068)	0.122	0.137	−0.331 (−0.665, 0.003)	0.053	0.087

(Continued)

Table 5. (Continued)

Proteins*	Univariable Regression			Multivariable Regression		
	β (95% CI) †	P value	P _{FDR} value‡	β (95% CI)†	P value	P _{FDR} value‡
XG	−0.024 (−0.361, 0.314)	0.891	0.891	−0.461 (−0.931, 0.008)	0.055	0.088
CTSZ	−0.417 (−0.725, −0.109)	0.008	0.015	−0.303 (−0.613, 0.008)	0.056	0.088
UMOD	0.488 (0.171, 0.805)	0.003	0.005	0.322 (−0.009, 0.653)	0.057	0.088
FAM3C	−0.571 (−0.905, −0.236)	0.001	0.002	−0.337 (−0.694, 0.019)	0.065	0.092
CLEC1A	−0.421 (−0.754, −0.087)	0.014	0.022	−0.316 (−0.651, 0.019)	0.065	0.092
PRTN3	−0.319 (−0.648, 0.010)	0.058	0.075	−0.313 (−0.643, 0.016)	0.063	0.092
THBD	−0.563 (−0.915, −0.210)	0.002	0.004	−0.349 (−0.716, 0.017)	0.062	0.092
IL2RA	−0.445 (−0.781, −0.109)	0.010	0.016	−0.320 (−0.665, 0.024)	0.069	0.096
PRSS2	−0.337 (−0.618, −0.056)	0.019	0.028	−0.260 (−0.543, 0.024)	0.073	0.099
CLEC5A	−0.266 (−0.598, 0.067)	0.118	0.135	−0.315 (−0.671, 0.042)	0.085	0.113
ART3	−0.419 (−0.730, −0.109)	0.008	0.015	−0.293 (−0.630, 0.043)	0.088	0.115
RARRES2	−0.374 (−0.733, −0.015)	0.041	0.055	−0.284 (−0.653, 0.086)	0.133	0.169
CXCL8	−0.386 (−0.721, −0.051)	0.024	0.034	−0.266 (−0.612, 0.081)	0.133	0.169
IGFBP2	−0.288 (−0.587, 0.011)	0.059	0.075	−0.238 (−0.557, 0.080)	0.143	0.177
RETN	−0.256 (−0.573, 0.061)	0.114	0.132	−0.228 (−0.552, 0.096)	0.168	0.205
IL19	−0.300 (−0.638, 0.038)	0.083	0.101	−0.229 (−0.568, 0.110)	0.186	0.223
CTSL	−0.378 (−0.717, −0.040)	0.029	0.039	−0.223 (−0.570, 0.124)	0.208	0.246
MFAP5	−0.431 (−0.773, −0.090)	0.014	0.022	−0.226 (−0.582, 0.129)	0.213	0.248
PDGFRA	−0.279 (−0.617, 0.059)	0.107	0.126	−0.189 (−0.530, 0.151)	0.276	0.310
DKK3	−0.320 (−0.649, 0.010)	0.058	0.075	−0.192 (−0.533, 0.150)	0.272	0.310
REG1A	−0.300 (−0.634, 0.034)	0.079	0.098	−0.187 (−0.527, 0.153)	0.282	0.313
ROR1	−0.258 (−0.597, 0.081)	0.136	0.151	−0.184 (−0.534, 0.166)	0.304	0.331
REG3A	−0.193 (−0.524, 0.138)	0.253	0.272	−0.147 (−0.485, 0.192)	0.397	0.426
CA4	−0.128 (−0.422, 0.166)	0.394	0.411	−0.121 (−0.433, 0.191)	0.447	0.473
CCL27	−0.046 (−0.364, 0.272)	0.776	0.787	0.125 (−0.206, 0.455)	0.459	0.479
CLC	−0.218 (−0.521, 0.086)	0.161	0.175	−0.095 (−0.400, 0.210)	0.542	0.558
REG1B	−0.169 (−0.488, 0.150)	0.299	0.317	−0.035 (−0.360, 0.290)	0.832	0.844
GRK5	−0.073 (−0.394, 0.247)	0.654	0.673	−0.021 (−0.340, 0.298)	0.899	0.899

*Adjusted for age, sex, smoking, systolic blood pressure, HbA1c, and duration of diabetes.

†Per-SD change of retinal nerve fiber layer thickness.

‡Adjusted for multiple testing (Benjamini-Hochberg procedure).

CI, confidence interval.

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fully concordant with GDES-PPP findings, including key markers such as ACTA2, COL6A3, CD46, RNASET2, CCL15, NT-proBNP, NECTIN2, and HSPG2, with β values ranging from $-0.716 \mu\text{m}$ (95% CI $[-1.077, -0.355]$) to $-0.399 \mu\text{m}$ (95% CI $[-0.717, -0.082]$). UMOD maintained a protective trend ($\beta = 0.322$), though not reaching significance ($P = 0.088$). Core GDES protein signatures are reproducible across populations, with consistent effects and biological relevance, highlighting their robust translational potential for DRN.

We further performed model-level external validation in UKB-PPP using a cross-sectional thin-RNFL phenotype (thin-nest quartile of baseline pRNFL thickness). Adding Pro-DRN consistently improved discrimination across all conventional frameworks (e.g., Age and Sex: 0.635 to 0.714; All model: 0.676 to 0.756; [S11 Table](#)) and improved calibration as assessed by lower Brier scores and no evidence of lack of fit by the Hosmer–Lemeshow test ([S12 Table](#)).

Discussion

Over the past two decades, DRD has been reconceptualised from a purely microvascular disorder to a neurovascular unit (NVU) disease, in which DRN emerges as an early and independent driver of disease progression [3,50]. By integrating high-throughput plasma proteomics with longitudinal multimodal retinal imaging, we systematically identified 71 plasma proteins significantly associated with DRN, predominantly involved in inflammation and immune responses, extracellular matrix remodeling, and microcirculatory homeostasis. Leveraging these molecular features, we constructed an AI-based predictive model, Pro-DRN, which demonstrated excellent discrimination that further improved when combined with clinical variables, substantially outperforming conventional risk factors. External validation in the UK Biobank confirmed these associations, demonstrating robustness across populations. An online platform was developed to facilitate clinical translation, enabling precise early risk assessment and neuroprotective intervention before irreversible retinal damage occurs. To our knowledge, this is among the earliest multi-cohort efforts to demonstrate the predictive value of plasma proteomics for DRN.

Unlike earlier proteomic studies, which have focused largely on late-stage vascular lesions and have typically drawn on intraocular fluids or animal models, this study extends the field in three complementary ways [17,29–31,51–53]. First, it prospectively targets a longitudinal neurodegenerative phenotype (OCT-quantified RNFL thinning rate). Second, it applies large-scale plasma proteomics to detect systemic molecular signals amenable to clinical implementation. Third, it integrates a multi-algorithm ML framework to handle complex nonlinear relationships in high-dimensional data, uncovering subtle yet reproducible signals that traditional methods may miss. Taken together, these choices reposition DRN from a condition recognized late at the imaging level to one that can be stratified early and mechanism-guided at the molecular level.

The Pro-DRN model, trained exclusively on proteomic features, achieved a C-index of 0.860, outperforming conventional models based on clinical variables (C-index 0.50–0.69). Combining proteomic data with clinical parameters further enhanced predictive performance to a C-index of 0.908, highlighting the independent and additive value of proteomic biomarkers. These findings align with growing evidence that non-traditional biomarkers, particularly proteomic signatures, can improve risk stratification by detecting early pathophysiological perturbations, such as neurovascular unit dysfunction, before irreversible structural damage occurs. Comparable gains have been reported in proteomic prediction of diabetic nephropathy and retinopathy, where inflammation- and tissue-injury-related proteins consistently outperform routine clinical metrics. These results argue for incorporating plasma proteomic profiling into precision prognostics for diabetic neurodegenerative complications, enabling the early identification of high-risk individuals and the timely delivery of targeted interventions before overt disease progression.

ML and AI served a dual purpose in this work: it modeled the high-dimensional proteomic data and it stress-tested the robustness of the identified signals. We benchmarked eight algorithms spanning ensemble gradient boosting, bagging, neural networks, and classical linear and instance-based learners within a unified cross-validation framework. Despite their very different theoretical foundations, every algorithm yielded concordant gains in discrimination, indicating that the proteomic signal rather than any particular modeling choice drives the predictive performance of Pro-DRN. Tree-based models, particularly XGBoost, excelled at capturing complex relationships between proteomic and imaging features due to their capacity for high-order interactions and minimal distributional assumptions [54–57]. This algorithmic concordance establishes that the plasma proteomic signatures of DRN are stable and generalizable, providing a secure foundation for biological interpretation and clinical translation.

Cross-cohort external validation is indispensable for establishing the robustness of candidate biomarkers and the generalisability of risk prediction models [58]. Using the UKB-PPP cohort, which differs substantially in genetic background, lifestyle, and epidemiology, we successfully reproduced the associations of core plasma proteins with DRN. Both the univariate and the multivariable analyses recapitulated the direction and magnitude of association for key markers, including ACTA2, COL6A3, and HSPG2, indicating that these biomarkers reflect intrinsic biological features of DRN rather than population- or platform-specific phenomena.

To address the “black box” nature of AI-based predictions, we applied SHAP (Shapley Additive Explanations), which decomposes each prediction into quantitative, signed contributions from every protein. Proteins such as COL6A3, ACTA2, HSPG2, CCL16, REG3A, PAM, ART3, and TIMP1 jointly formed the primary positive risk contributions. These analyses revealed that the Pro-DRN score is governed by the cumulative and interactive contributions of multiple proteins rather than by any single dominant marker. Integrating these findings with functional enrichment analyses, the signals can be grouped into four core pathological axes: (1) inflammation and immune recruitment (e.g., CCL16), (2) basement membrane and extracellular matrix remodeling (e.g., COL6A3, HSPG2), (3) membrane-receptor function and blood–retina barrier (BRB) integrity (e.g., HSPG2), and (4) microcirculatory homeostasis (e.g., ACTA2). These pathways align with prior hypotheses regarding pericyte loss, BRB disruption, and glial activation, collectively highlighting NVU instability as a central molecular driver of early DRN [1,3,7,59,60].

Key proteins illuminate DRN pathology. Elevated ACTA2 signals phenotypic transformation of retinal pericytes or smooth muscle-like cells, with ischemic stress triggering sustained contraction, capillary spasm, hypoperfusion, axonal transport disruption, metabolic stress, and glial activation, accelerating neurodegeneration [61]. COL6A3, a major extracellular matrix and basement membrane protein, undergoes diabetic remodeling that alters matrix mechanics, permeability, bioactive factor release, and axonal support, destabilizing the neuronal microenvironment [62,63]. HSPG2 regulates vascular permeability and barrier integrity, linking circulating protein signals to RNFL thinning [64]. Collectively, microvascular dysfunction, matrix remodeling, and BRB disruption form an interconnected network whose feedback loops drive DRN through NVU instability.

Among the candidate proteins, uromodulin (UMOD) stood out for its protective association with retinal nerve fiber layer preservation. Classically regarded as a kidney-derived protein secreted by distal tubular epithelial cells, UMOD is known to regulate ion transport, innate immunity, and epithelial barrier function [65,66]. Our findings implicate UMOD in DRN and suggest a “kidney–vasculature–retina axis” that modulates retinal neuronal vulnerability [67,68]. In diabetes, renal impairment amplifies inflammation, metabolic dysregulation, and endothelial dysfunction, destabilizing the NVU and accelerating glial activation and microcirculatory injury [69]. The protective association observed here likely reflects the systemic homeostatic role of healthy renal tubules, which confers resistance to the inflammatory cascades and endothelial dysfunction that precipitate neurovascular injury. Clinically, diabetic nephropathy and retinopathy often coexist and correlate in severity, suggesting that preserving tubular health may also protect retinal neurons, offering a cross-organ strategy for early prevention of diabetic complications [68,70–72].

The retina, as an embryological outgrowth of the diencephalon, offers an accessible window onto central nervous system biology. Substantial evidence indicates that diabetic patients face elevated risks of cognitive impairment and Alzheimer’s disease (AD), with early pathological changes often mirrored in the retina [73–76]. For example, inner retinal thinning correlates with mild cognitive impairment, and multifocal ERG abnormalities can predict future cognitive decline [9,10,77,78]. Notably, several DRN-risk proteins identified here (e.g., COL6A3, HSPG2, TIMP1) have also been implicated in AD and Parkinson’s disease, suggesting shared molecular underpinnings between retinal and central neurodegenerative disorders [14,15,20,22,79,80]. DRN should therefore be considered not merely as a localized complication of diabetes but as a potential early biomarker of systemic neurodegeneration. Future studies should prospectively evaluate whether these protein markers jointly predict DRN and cognitive decline, enabling development of blood-based, cross-disease early-warning systems [18].

These findings carry several clinical translational implications. First, Pro-DRN provides a quantitative tool for early risk stratification, enabling efficient allocation of limited ophthalmic screening and surveillance resources to those who stand to benefit most. High-risk individuals could benefit from intensified surveillance, such as shorter OCT follow-up intervals, and early interventions that complement current guidelines focused on vascular lesions while often neglecting neurodegeneration [45,81–83]. Second, it offers an objective enrichment criterion for selecting participants in neuroprotective trials, including trials of GLP-1 receptor agonists, DPP-4 inhibitors, and pathway-specific investigational agents, which should substantially enhance trial efficiency and statistical power [84,85]. Third, the proteins and pathways implicated by our analyses constitute a set of human-omics-derived candidate targets for the development of novel therapies for DRN.

The strengths of this study include its large-scale prospective design, six-year longitudinal OCT phenotyping, high-throughput plasma proteomics, and rigorous cross-cohort validation. Several limitations should nonetheless be acknowledged. First, circulating plasma protein levels do not fully capture local retinal expression and may therefore incompletely reflect the pathogenic mechanisms operating in retinal tissue. Future studies should integrate spatial proteomics of ocular tissues and employ cellular or animal models to dissect the specific roles and therapeutic potential of candidate molecules in DRN. Second, the RNFL measurement region differed between cohorts, being peripapillary in GDES and macular in UK Biobank; although prior evidence indicates that both regions reliably reflect neurodegeneration, residual non-comparability cannot be excluded. Third, our proteomic measurements were obtained at a single time point, precluding any evaluation of temporal proteomic trajectories or their dynamic relationship with disease progression. Cost-efficient, targeted panels could enable large-scale longitudinal monitoring in future studies. Fourth, while Pro-DRN achieved strong discrimination, predictive performance might be further enhanced by integrating additional layers of data, including genomic risk scores, epigenomic features, wearable-sensor metrics, and environmental exposures. Finally, Olink targeted proteomics, while broad, samples only a subset of the human proteome, so that potentially informative proteins and mechanisms remain unexplored. Advances in ultra-high-throughput proteomics and AI promise more comprehensive molecular characterization of DRN [15,16,18].

In summary, by integrating high-throughput plasma proteomics with multi-algorithm ML and explainable AI, this multi-cohort study identified a robust, cross-population plasma proteomic signature of early DRN. The resulting Pro-DRN model achieved high predictive performance, supported excellent calibration and clinical utility, and translates seamlessly into a deployable online decision-support tool, together marking a shift from late imaging-based recognition to early molecular prediction of DRN. Beyond prediction, these findings illuminate the molecular architecture of neurovascular unit instability in early diabetes and establish a foundation for high-risk screening, neuroprotective trial enrichment, and the rational design of targeted therapies.

Ethics approval

The study was approved by the Northwest Multicenter Research Ethics Committee (11/NW/0382) and the Ethics Committee of Zhongshan Ophthalmic Center (2017KYPJ094).

Supporting information

S1 Checklist. STROBE statement. Based on the STROBE Statement (Strengthening the Reporting of Observational Studies in Epidemiology). Available from: <https://www.strobe-statement.org/>. The STROBE Statement is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0).
(DOCX)

S2 Checklist. TRIPOD checklist. Based on the TRIPOD Statement (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis). Available from: <https://www.tripod-statement.org/>. The TRIPOD Statement is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0).
(DOCX)

S1 Text. Supplementary methods.
(DOCX)

S1 Table. Baseline characteristics of the population 1 (GDES-PPP cross-sectional cohort) and population 3 (UKB-PPP validation cohort).
(DOCX)

S2 Table. Proteins associated with average retinal nerve fiber layer thickness.
(DOCX)

S3 Table. Proteins associated with the thinning rate of retinal nerve fiber layer thickness.

(DOCX)

S4 Table. Proteins associated with the thinning rate of retinal nerve fiber layer thickness in the sensitivity analysis with further adjustment for baseline retinal nerve fiber layer thickness.

(DOCX)

S5 Table. Performance of Pro-DRN for predicting DRN with various ML algorithms in the GDES cohort.

(DOCX)

S6 Table. Comparison of the performance between conventional predictors and Pro-DRN for predicting DRN.

(DOCX)

S7 Table. Comparison of the performance of incorporating Pro-DRN with the full proteins in the GDES.

(DOCX)

S8 Table. Incremental predictive value of Pro-DRN beyond metabolically augmented conventional clinical models.

(DOCX)

S9 Table. Discrimination of Pro-DRN under alternative DRN definitions.

(DOCX)

S10 Table. Incremental discrimination of Pro-DRN beyond conventional clinical frameworks under alternative DRN definitions.

(DOCX)

S11 Table. Model-level external validation of Pro-DRN in UKB-PPP using a cross-sectional thin-RNFL outcome.

(DOCX)

S12 Table. Assessment of model calibration for Pro-DRN in UKB-PPP using a cross-sectional thin-RNFL outcome.

(DOCX)

S13 Table. Predictors used in conventional prediction models.

(DOCX)

S14 Table. Protein names and gene names for proteins identified in this study.

(DOCX)

S1 Data. Supplementary data for [Figs 2](#), [3](#), and [5](#).

(XLSX)

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